

Étale Coverings and the Fundamental Group

(SGA 1)

English source-synchronization workpass

Exposé I opening through § I.5

Machine-assisted bounded working translation

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Source and status note

This bounded workpass translates the opening of Exposé I through §I.5 directly from the French $\mathrm{T}_{\mathrm{E}}\mathrm{X}$ source in arXiv `math/0206203v2`, with archive SHA-256:

0A33C9C06908705A2525690FAAE02F6F
07980A4AA069A8B6CFF9B1D9BC39ACD3

The current translated source boundary is `smf_doc-math_3_01.tex`, lines 556–1167; line 1168 begins §I.6 and is outside the current workpass. The opening through §I.4 is sealed in the separately frozen `r3` checkpoint and is preserved as prior evidence; publication acceptance and readback are not claimed here. The §I.5 addition is the current source-, formula-, structure-, terminology-, build-, and render-review unit.

Recovered English materials associated with `jcreinhold`, `jmoellermath`, and Tim Hosgood were consulted only as target-language comparison controls. They are not source authority, and no agreement among them was treated as proof of source alignment. The two Hosgood-related repositories belong to one derivative lineage, not two independent witnesses.

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Contents

Source and status note	i
Chapter I. Étale morphisms	1
I.1. Notions of differential calculus	1
I.2. Quasi-finite morphisms	1
I.3. Unramified or net morphisms	1
I.4. Étale morphisms. Étale coverings	3
I.5. The fundamental property of étale morphisms	4

CHAPTER I

Étale morphisms

To simplify the exposition, we assume that all preschemes under consideration are locally noetherian, at least after no. I.2.

French source
p. 1

I.1. Notions of differential calculus

Let X be a prescheme over Y , and let $\Delta_{X/Y}$, or simply Δ , be the diagonal morphism $X \rightarrow X \times_Y X$. It is an immersion, hence a closed immersion of X into an open subset V of $X \times_Y X$. Let $\mathcal{I}_X$ be the ideal of the closed subscheme corresponding to the diagonal in V (N.B. if one wishes to proceed intrinsically, without assuming that X is separated over Y —a hypothesis that would be farcical—one should consider the set-theoretic inverse image of $\mathcal{O}_{X \times X}$ in X , and denote by $\mathcal{I}_X$ the augmentation ideal in the latter...). The sheaf $\mathcal{I}_X/\mathcal{I}_X^2$ may be regarded as a quasi-coherent sheaf on X ; we denote it by $\Omega_{X/Y}^1$. It is of finite type if $X \rightarrow Y$ is of finite type. It behaves well under base change $Y' \rightarrow Y$. We also introduce the sheaves $\mathcal{O}_{X \times_Y X}/\mathcal{I}_X^{n+1} = \mathcal{P}_{X/Y}^n$; these are sheaves of *rings* on X , making X into a prescheme that may be denoted by $\Delta_{X/Y}^n$ and called the *n-th infinitesimal neighborhood of X/Y* . The resulting sorites is entirely trivial, although rather long¹; it would be prudent to speak of it only when there is something useful to say about it, in connection with smooth morphisms.

I.2. Quasi-finite morphisms

PROPOSITION I.2.1. *Let $A \rightarrow B$ be a local homomorphism (N.B. the rings are now noetherian), and let $\mathfrak{m}$ be the maximal ideal of A . The following conditions are equivalent:*

- (i) $B/\mathfrak{m}B$ is finite-dimensional over $k = A/\mathfrak{m}$.
- (ii) $\mathfrak{m}B$ is an ideal of definition, and $B/\mathfrak{r}(B) = \kappa(B)$ is an extension of $k = \kappa(A)$.
- (iii) The completion $\hat{B}$ is finite over the completion $\hat{A}$ of A .

One then says that B is *quasi-finite* over A . A morphism $f: X \rightarrow Y$ is said to be quasi-finite at x (or the Y -prescheme f is said to be quasi-finite at x) if $\mathcal{O}_x$ is quasi-finite over $\mathcal{O}_{f(x)}$. This is also equivalent to saying that x is *isolated in its fiber* $f^{-1}(f(x))$.² A morphism is said to be quasi-finite if it is so at every point.³

French source
p. 2

COROLLARY I.2.2. *If A is complete, quasi-finite is equivalent to finite.*

One could give the usual chain of implications (i), (ii), (iii), (iv), (v) for quasi-finite morphisms, but this does not seem indispensable here.

I.3. Unramified or net morphisms

PROPOSITION I.3.1. *Let $f: X \rightarrow Y$ be a morphism of finite type, let $x \in X$, and put $y = f(x)$. The following conditions are equivalent:*

¹Cf. EGA IV 16.3.

²The original printing (Exposé I, p. 2) and both French arXiv renderings print $f^{-1}(x)$. Since $x \in X$ for $f: X \rightarrow Y$, the type-correct fiber through x is $f^{-1}(f(x))$, which is used here with the source defect explicitly disclosed.

³In EGA II 6.2.3 one assumes in addition that f is of finite type.

- (i) $\mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x$ is a finite separable extension of $\kappa(y)$.
- (ii) $\Omega_{X/Y}^1$ vanishes at x .
- (iii) The diagonal morphism $\Delta_{X/Y}$ is an open immersion in a neighborhood of x .

For the implication (i) $\Rightarrow$ (ii), Nakayama's lemma immediately reduces us to the case $Y = \operatorname{Spec}(k)$, $X = \operatorname{Spec}(k')$, where the assertion is well known and, moreover, immediate from the definition of separability. The implication (ii) $\Rightarrow$ (iii) follows from a pleasant and easy characterization of open immersions using Krull. For (iii) $\Rightarrow$ (i), one is again reduced to the case $Y = \operatorname{Spec}(k)$, with the diagonal morphism an open immersion everywhere. One must then prove that X is finite, with a separable coordinate ring over k ; for this one reduces to the case where k is algebraically closed. But then every closed point of X is isolated (it is precisely the inverse image of the diagonal under the morphism $X \rightarrow X \times_k X$ defined by x), and hence X is finite. One may therefore suppose that X consists of a single point, with ring A . Thus $A \otimes_k A \rightarrow A$ is an isomorphism, whence $A = k$, as required.

- DEFINITION I.3.2. (a) One then says that f is *net*, or *equivalently unramified*, at x , or that X is *net*, or *unramified*, at x over Y .
- (b) Let $A \rightarrow B$ be a local homomorphism. It is said to be *net*, or *unramified*, and B is said to be a *net*, or *unramified*, local algebra over A , if $B/\mathfrak{r}(A)B$ is a finite separable extension of $A/\mathfrak{r}(A)$; that is, if $\mathfrak{r}(A)B = \mathfrak{r}(B)$ and $\kappa(B)$ is a separable extension of $\kappa(A)$.⁴

Remarks. The fact that B is unramified over A can already be checked on the completions of A and B . Unramified implies quasi-finite.

French source
p. 3

COROLLARY I.3.3. The set of points at which f is unramified is open.

COROLLARY I.3.4. Let X' and X be two preschemes of finite type over Y , and let $g: X' \rightarrow X$ be a Y -morphism. If X is unramified over Y , the graph morphism

$$\Gamma_g: X' \longrightarrow X' \times_Y X$$

is an open immersion.⁵

Indeed, it is the inverse image of the diagonal morphism $X \rightarrow X \times_Y X$ under

$$g \times_Y \operatorname{id}_X: X' \times_Y X \longrightarrow X \times_Y X.$$

One may also introduce the annihilator ideal $\mathfrak{d}_{X/Y}$ of $\Omega_{X/Y}^1$, called the *different ideal* of X/Y . It defines a closed subprescheme of X whose underlying set is the set of points at which X/Y is ramified, that is, not unramified.

- PROPOSITION I.3.5. (i) An immersion is unramified.
- (ii) The composite of two unramified morphisms is unramified.
- (iii) A base change of an unramified morphism is unramified.

This may be seen equally well from criterion (ii) or criterion (iii) (the latter seems more amusing to me). One can of course also be more precise by giving pointwise statements; this is more general only in appearance (except in the setting of Definition I.3.2(b)), and tedious. As usual, one obtains the following corollaries.

- COROLLARIES I.3.6. (iv) The Cartesian product of two unramified morphisms is unramified.
- (v) If gf is unramified, then f is unramified.

⁴Cf. the regrets in Exposé III, no. 1.2.

⁵The original printing (Exposé I, p. 3) and the French TeX give $X \times_Y X$ as the graph target and later write $\operatorname{id}_{X'}$ in the pullback map. The typed fiber products force $X' \times_Y X$ and id_X , respectively; those type-correct readings are used here with both source defects disclosed.

(vi) If f is unramified, then f_{red} is unramified.

PROPOSITION I.3.7. *Let $A \rightarrow B$ be a local homomorphism. Suppose that the residue-field extension $\kappa(B)/\kappa(A)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is unramified if and only if $\hat{B}$, as an $\hat{A}$ -algebra, is a quotient of $\hat{A}$.*

Remarks.

- When the residue-field extension is not assumed to be trivial, one may reduce to that case by making a suitable finite flat extension over A that kills the extension in question.
- Take the example in which A is the local ring of an ordinary double point of a curve and B is the local ring at a point of the normalization. Then $A \subset B$, and B is unramified over A with trivial residue-field extension; the map $\hat{A} \rightarrow \hat{B}$ is surjective but *not injective*. We shall therefore strengthen the notion of unramifiedness.

French source
p. 4

I.4. Étale morphisms. Étale coverings

We shall take for granted whatever is needed concerning flat morphisms; these facts will be proved later, if necessary⁶.

- DEFINITION I.4.1. (a) *Let $f: X \rightarrow Y$ be a morphism of finite type. One says that f is étale at x if f is flat at x and net (equivalently, unramified) at x . One says that f is étale if it is so at every point. One says that X is étale at x over Y , or that it is a Y -prescheme étale at x , and so on.*
- (b) *Let $f: A \rightarrow B$ be a local homomorphism. One says that f is étale, or that B is étale over A , if B is flat and unramified over A .⁷*

PROPOSITION I.4.2. *For B/A to be étale, it is necessary and sufficient that $\hat{B}/\hat{A}$ be étale.*

Indeed, this holds separately for “unramified” and for “flat.”

COROLLARY I.4.3. *Let $f: X \rightarrow Y$ be of finite type, and let $x \in X$. Whether f is étale at x depends only on the local homomorphism $\mathcal{O}_{f(x)} \rightarrow \mathcal{O}_x$, and even only on the corresponding homomorphism between the completions.*

COROLLARY I.4.4. *Suppose that the residue-field extension $\kappa(A) \rightarrow \kappa(B)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is étale if and only if $\hat{A} \rightarrow \hat{B}$ is an isomorphism.*

This follows by combining flatness with Proposition I.3.7.

PROPOSITION I.4.5. *Let $f: X \rightarrow Y$ be a morphism of finite type. Then the set of points at which it is étale is open.*

Indeed, this holds separately for “unramified” and for “flat.”

This proposition shows that one may dispense with “pointwise” statements when studying morphisms of finite type that are étale at some point.

- PROPOSITION I.4.6. (i) *An open immersion is étale.*
- (ii) *The composite of two étale morphisms is étale.*
- (iii) *A base change of an étale morphism is étale.*

Indeed, (i) is trivial, and for (ii) and (iii) it is enough to note that the assertions hold for “unramified” and for “flat.” To tell the truth, there are also corresponding statements for local homomorphisms (without a finiteness condition); in any event, they will have to appear in the *multiplodoque*, beginning with the unramified case.

French source
p. 5

⁶Cf. Exposé IV.

⁷Cf. the regrets in Exposé III, no. 1.2.

COROLLARY I.4.7. *The Cartesian product of two étale morphisms is likewise étale.*

COROLLARY I.4.8. *Let X and X' be of finite type over Y , and let $g: X \rightarrow X'$ be a Y -morphism. If X' is unramified over Y and X is étale over Y , then g is étale.*

Indeed, g is the composite of the graph morphism

$$\Gamma_g: X \longrightarrow X \times_Y X',$$

which is an open immersion by Corollary I.3.4, and the projection $X \times_Y X' \rightarrow X'$, which is étale because it is obtained from the étale morphism $X \rightarrow Y$ by the base change $X' \rightarrow Y$.

DEFINITION I.4.9. *An étale covering (respectively, a net covering, also called an unramified covering) of Y is a Y -scheme X that is finite over Y and étale (respectively, unramified) over Y .*

The first condition means that X is defined by a coherent sheaf of algebras $\mathcal{B}$ on Y . The second then means that $\mathcal{B}$ is locally free over Y (with no corresponding local-freeness requirement in the net case), and, moreover, that for every $y \in Y$ the fiber

$$\mathcal{B}(y) = \mathcal{B}_y \otimes_{\mathcal{O}_y} \kappa(y)$$

is a separable algebra over $\kappa(y)$, that is, a finite product of finite separable field extensions of $\kappa(y)$.

PROPOSITION I.4.10. *Let X be a flat covering of Y of degree n (the definition of this term deserved to appear in Definition I.4.9), defined by a coherent locally free sheaf of algebras $\mathcal{B}$. In the usual way one defines the trace homomorphism $\mathcal{B} \rightarrow \mathcal{A}$, which is a homomorphism of $\mathcal{A}$ -modules, where $\mathcal{A} = \mathcal{O}_Y$. For X to be étale, it is necessary and sufficient that the corresponding trace pairing $(x, y) \mapsto \text{Tr}_{\mathcal{B}/\mathcal{A}}(xy)$ induce an isomorphism $\mathcal{B} \xrightarrow{\sim} \mathcal{B}^\vee$,⁸ or, equivalently, that the discriminant section*

$$d_{X/Y} = d_{\mathcal{B}/\mathcal{A}} \in \Gamma\left(Y, \bigwedge^n \mathcal{B}^\vee \otimes_{\mathcal{A}} \bigwedge^n \mathcal{B}^\vee\right)$$

be invertible, or, finally, that the discriminant ideal defined by this section be the unit ideal.

Indeed, one reduces to the case $Y = \text{Spec}(k)$; the assertion is then the well-known separability criterion, and becomes immediate after passing to an algebraic closure of k .

Remark. There will be a less trivial statement below, when one does not assume a priori that X is flat over Y , but instead imposes a normality hypothesis.

French source
p. 6

I.5. The fundamental property of étale morphisms

THEOREM I.5.1. *Let $f: X \rightarrow Y$ be a morphism of finite type. For f to be an open immersion, it is necessary and sufficient that it be étale and radicial.*

Recall that radicial means injective, with radicial residue-field extensions; one may also recall that this means that the morphism remains injective after every base change. Necessity is trivial; it remains to prove sufficiency. We shall give two different proofs, the first shorter and the second more elementary.

- 1) A flat morphism is open, so (replacing Y by $f(X)$) we may suppose that f is a homeomorphism onto Y . After every base change, f remains flat, radicial, and surjective, hence a homeomorphism and therefore closed. Thus f is proper. Hence f is finite (reference: Chevalley's theorem) and is defined by a coherent sheaf $\mathcal{B}$ of algebras. The sheaf $\mathcal{B}$ is locally free and, by the hypothesis, has rank 1 everywhere. Thus $X = Y$, as required.

⁸The French TeX and the original printing instead give $\mathcal{B}$ as the target. An $\mathcal{A}$ -bilinear form induces a map $\mathcal{B} \rightarrow \mathcal{B}^\vee$; the dual is therefore the type-correct target, used here with the source-original defect disclosed.

- 2) One may suppose Y and X *affine*. Moreover, one easily reduces to proving the following: if $Y = \operatorname{Spec}(A)$, with A local, and if $f^{-1}(y)$ is nonempty, where y is the closed point of Y , then $X = Y$. Indeed, this will imply that every $y \in f(X)$ has an open neighborhood U such that $X|_U = U$. We shall have $X = \operatorname{Spec}(B)$, and we want to prove $A = B$. For this, it is enough to prove the analogous assertion after replacing A by $\hat{A}$ and B by $B \otimes_A \hat{A}$, taking into account that $\hat{A}$ is faithfully flat over A . We may therefore suppose that A is *complete*. Let x be the point above y . By Corollary I.2.2, $\mathcal{O}_x$ is finite over A , and hence, being flat and radicial over A , is A itself. Thus

$$X = Y \amalg X'$$

as a disjoint union. Since X is radicial over Y , X' is empty. This completes the proof.

COROLLARY I.5.2. *Let $f: X \rightarrow Y$ be both a closed immersion and étale. If X is connected, f is an isomorphism from X onto a connected component of Y .*

Indeed, f is also an open immersion. It follows that:

French source
p. 7

COROLLARY I.5.3. *Let X be an unramified Y -scheme, with Y connected. Then every section of X over Y is an isomorphism from Y onto a connected component of X . There is therefore a one-to-one correspondence between the set of such sections and the set of connected components X_i of X for which the projection $X_i \rightarrow Y$ is an isomorphism (or, equivalently by Theorem I.5.1, is surjective, étale, and radicial).⁹ In particular, a section is determined by its value at one point.*

Only the first assertion requires proof. By Corollary I.5.2, it is enough to observe that a section is a closed immersion (because X is separated over Y) and is étale by Corollary I.4.8.

COROLLARY I.5.4. *Let X and Y be preschemes over S , with X unramified and separated over S and Y connected. Let $f, g: Y \rightarrow X$ be two S -morphisms, and let $y \in Y$. Suppose that $f(y) = g(y) = x$ and that the homomorphisms of residue fields $\kappa(x) \rightarrow \kappa(y)$ induced by f and g are identical (“ f and g coincide geometrically at y ”). Then f and g are identical.*

This follows from Corollary I.5.3 by reducing to the case $Y = S$, replacing X by $X \times_S Y$.

Here is a particularly important variant of Corollary I.5.3.

THEOREM I.5.5. *Let S be a prescheme, let X and Y be two S -preschemes, and let S_0 be a closed subscheme of S having the same underlying space as S . Put*

$$X_0 = X \times_S S_0 \quad \text{and} \quad Y_0 = Y \times_S S_0,$$

the “restrictions” of X and Y to S_0 . Suppose that X is étale over S . Then the natural map

$$\operatorname{Hom}_S(Y, X) \longrightarrow \operatorname{Hom}_{S_0}(Y_0, X_0)$$

is bijective.

We are again reduced to the case $Y = S$; the assertion then follows from the “topological” description of the sections of X/Y given in Corollary I.5.3.

⁹The French TeX and the original printing give only “surjective and radicial” and cite Theorem I.5.1, which also requires étaleness. Without it the claim is false: for example, $\operatorname{Spec}(k) \rightarrow \operatorname{Spec}(k[\varepsilon]/(\varepsilon^2))$ is surjective on points, radicial, unramified, separated, and of finite type, but is not an isomorphism. The logically correct criterion is used here with the source defect disclosed.

Scholium. This result includes both a *uniqueness* assertion and an *existence* assertion for *morphisms*. It may also be expressed, when X and Y are both taken to be étale over S , by saying that the functor

$$X \longmapsto X_0$$

from the category of étale S -schemes to the category of étale S_0 -schemes is *fully faithful*; that is, it establishes an *equivalence* of the first category with a *full subcategory* of the second. We shall see below that it is actually an equivalence between the first and the second (which will be an *existence theorem for étale S -schemes*).

The following form of Theorem I.5.5, apparently more general, is often convenient.

COROLLARY I.5.6 (“Extension theorem for liftings”). *Consider a commutative diagram of morphisms*

$$\begin{array}{ccc} X & \longleftarrow & Y_0 \\ \downarrow & & \downarrow \\ S & \longleftarrow & Y \end{array}$$

in which $X \rightarrow S$ is étale and $Y_0 \rightarrow Y$ is a bijective closed immersion. Then there is a unique morphism $Y \rightarrow X$ that makes the two corresponding triangles commute.

Indeed, replacing S by Y and X by $X \times_S Y$, we are reduced to the case $Y = S$; this is then the special case of Theorem I.5.5 with $Y = S$.

We should also note the following immediate consequence of Theorem I.5.1 (which we did not state as Corollary 1, so as not to interrupt the line of ideas developed after Theorem I.5.1):

PROPOSITION I.5.7. *Let X and X' be two preschemes of finite type and flat over Y , and let $g: X \rightarrow X'$ be a Y -morphism. For g to be an open immersion (respectively, an isomorphism), it is necessary and sufficient that, for every $y \in Y$, the induced morphism on the fibers*

$$g \otimes_Y \kappa(y): X \otimes_Y \kappa(y) \longrightarrow X' \otimes_Y \kappa(y)$$

be so.

It is enough to prove sufficiency. Since the assertion holds for surjectivity, we are reduced to the case of an open immersion. By Theorem I.5.1, one must verify that g is *radicial*, which is immediate, and that it is *étale*, which follows from Corollary I.5.9 below.

COROLLARY I.5.8. *(This should be moved to no. I.3.) Let X and X' be two preschemes over Y , let $g: X \rightarrow X'$ be a Y -morphism, let $x \in X$, and let y be its image in Y . For g to be quasi-finite (respectively, unramified) at x , it is necessary and sufficient that the same be true of $g \otimes_Y \kappa(y)$.*

Indeed, the two algebras over $\kappa(g(x))$ that one must examine to verify that the morphism is quasi-finite (respectively, unramified) at x are the same for g and for $g \otimes_Y \kappa(y)$.

COROLLARY I.5.9. *With the notation of Corollary I.5.8, suppose that X and X' are flat and of finite type over Y . For g to be flat (respectively, étale) at x , it is necessary and sufficient that $g \otimes_Y \kappa(y)$ be so.*

For “flat,” the statement is included only as a reminder: it is one of the fundamental criteria for flatness.¹⁰ For “étale,” the assertion follows from this, taking Corollary I.5.8 into account.

¹⁰Cf. Exposé IV, no. 5.9.